

CS 3120: Discrete Math and Theory 2 2026

Summer

MTWRF 10:30 am–12:45 pm in
Rice Hall 340
Instructor: Mark Floryan
Email: mfloryan@cs.virginia.edu
Office: Rice Hall 203
Office Hours: See course website

Teaching Assistants: See course website. Office hours will be held in-person. There may be some online office hours depending on the number of teaching assistants and the preferences of the students and course staff.

Course website: <https://markfloryan.github.io/dmt2/> (We will also use Canvas.)

Prerequisites: CS 4102 (old curriculum) or CS 3100 (new curriculum) with grades of C- or higher (Prerequisites are important to this course and will be enforced!)

Overview

Course Description: The goal of this course is to understand the fundamental limits on what can be efficiently computed. These limits reveal properties about information, communication, and computing, as well as practical issues about how to solve problems. Introduces computation theory including grammars, automata, and Turing machines.

Availability: It is important to us to be available to our students, and to address their concerns. If you cannot meet with us during our office hours, e-mail us and we will find the time to meet. That being said, like everybody else we are quite busy, so it may take a day or more to find a time to meet. And if you have any comments on the course—what is working, what is not working, what can be done better, etc.—we are very interested in hearing about them. Please send Prof. Floryan or one of the TAs an e-mail or post privately on Piazza to the instructors. When sending email, include “CS3120” in the subject line. If your question could be answered by either professor or even a TA, a post on Piazza to “instructors” may get a faster response.

Course Objectives: Students who complete the course will:

- Improve their mathematical thinking skill and habits, including thinking precisely about definitions, stating assumptions carefully, critically reading arguments, and being able to write convincingly.
- Be able to understand both finite and infinite formal models of computation and to reason about what they can and cannot compute.

- Understand both intuitively and formally what makes some problems too expensive to solve, and what can be done in practice when an unsolvable or intractable problem is encountered.
- Reason formally about the cost of computation, and be able to prove useful bounds on the costs of solving problems, including showing that certain problems are intractable.
- Learn about some interesting aspects of theoretical computer science, including cryptography and machine learning.

Textbook: *Introduction to the Theory of Computation, Second Edition* by Michael Sipser.

Additional Resources: We will make additional optional textbooks and resources available on the course website.

Class Delivery and Covid-19:

Lectures and quizzes will be given in-person and attendance is required.

Coursework and Grading

The course is divided into 4 **modules** that build upon one another:

- Introduction, Cardinality, Regular Languages
- Context-Free Grammars
- Turing Machines and Reducibility
- Complexity Theory

The modules are an average of 2-4 weeks worth of content. The schedule is shown on the course website.

Exams:

There will be three exams (two midterms and one final). The first exam will cover module one, the second will cover modules 2 and 3. The final exam will be cumulative with a focus on the final module. The number of exams and topic coverage of those exams is subject to change if necessary but will be announced clearly in class if so. The quiz dates are:

First exam (Module 1): Fri. July 24
Second exam (Modules 2-3): Fri. July 31
Final exam (Cumulative): Fri. Aug. 7

Final Exam: The final exam is scheduled for the last Friday of class (the final class period). Note that this is different from the University schedule, which has the final exam on Saturday Aug. 8.

Homeworks: Most of our homework assignments will be “written problem sets” with the occasional programming assignment. These assignments may include small problems, run-time analysis, proofs, etc. See section about L^AT_EX below. Handwritten assignments will NOT be accepted. This includes L^AT_EX documents that embed images of formulas or hand-drawn work in any way.

Grading: Your letter grade will be calculated using a traditional weighted average. The weights are:

- **Homework:** 20 percent (about 6 homeworks)
- **Attendance:** 20 percent (4 percent deduction for each missed lecture)
- **Midterm Exams:** 30 percent (15 percent each)
- **Final Exam:** 30 percent

Homeworks will be graded on a traditional percentage scale. Quizzes will be graded on a traditional percentage scale as well.

L^AT_EX: Written assignments must be typeset with L^AT_EX, a professional formatting system. Tutorials on how to use L^AT_EX will be made available when the first written problem set is released. L^AT_EX is easily installable on many computers:

- Overleaf, <http://overleaf.com>: a Web-hosted L^AT_EX editor which behaves much like Google Docs.
- Cygwin has L^AT_EX packages that can be installed
- MiKTeX provides a stand-alone installer for Windows and Mac, miktex.org
- Ubuntu and CentOS provide TeXLive packages in their repos
- LyX, TexShop, TeXworks, and TeXStudio are GUI editors available either through the MiKTeX and TeXLive repos or available as separate downloads.

We strongly recommend using Overleaf, <http://overleaf.com>, since it contains all the necessary packages and works in-browser. We generally will not accept L^AT_EX documents with images of text or formulas; **you must typeset the formulas in L^AT_EX**, not in another program and have them exported as images. You may also **NOT embed images of any handwritten work!**

Submission System: All homeworks will be submitted via GradeScope. Details will be explained later in the course.

Homework Late Policy: For homeworks, there will be a traditional due date, and no work will be accepted after this date. However, an extension request can be made by filling out an online form (link will be provided to students on course website). This request will enable students to earn a 3 day extension. No extensions will be given for any reason beyond this 3 day extension. In addition, extension requests **MUST** be submitted within 24 hours of the original assignment deadline. Failure to submit an extension request in time will result in the extension request being rejected.

Regrades: There will be a process for requesting regrades on assignments. This policy will be communicated once graded material is returned. Regrades will be submitted through Gradescope.

Collaboration Policy

Midterms and Exams: Exams are always individual work; collaboration with others is not allowed. It is also prohibited to use phones, smart-watches, outside notes, or any other assistance during an exam or quiz.

Homeworks: You are encouraged to collaborate with up to 2 other students in the course on each homework, and you may submit a shared set of solutions online. You are expected to collaborate on the work together, and solutions from homework assignments will appear on exams to ensure you are doing so.

On programming assignments, you can discuss the problem with up to 2 other students, but must work on your implementation individually. In addition, you will submit the assignment individually. In other words, you may not look at another student's code for any reason for programming assignments.

Plagiarism, though, is strictly not allowed and will result in a penalty (see below).

Penalty: Assignments or quizzes where violations of this policy occur will receive a penalty of a **zero on the assignment in question** plus a **zero on one equally weighted assignment (i.e., a second quiz or second homework)** and **one letter grade (e.g., A- drops to a B+)** on the student's **Final course grade** for the first offense. For second offenses, the student's **Final course grade** will be set to an **F** and a reference to the honor committee will be made. *Please note that this policy is a guideline and the instructors are entitled to levy a different penalty on a case-by-case basis, particularity of the severity of the infraction is extreme.*

Additional Information

Special Circumstances: The University of Virginia strives to provide accessibility to all students. If you require an accommodation to fully access this course, please contact the Student Disability Access Center (SDAC) at (434) 243-5180 or sdac@virginia.edu. If you are unsure if you require an accommodation, or to learn more about their services, you may contact the SDAC at the number above or by visiting their website <http://studenthealth.virginia.edu/sdac>.

For this course, we ask that students with special circumstances let us know as soon as possible, preferably during the **first week of class**.

Religious Accommodations: It is the University's long-standing policy and practice to reasonably accommodate students so that they do not experience an adverse academic consequence when sincerely held religious beliefs or observances conflict with academic requirements. Students who wish to request academic accommodation for a religious observance

should submit their request in writing to Prof. Floryan as far in advance as possible. If you have questions or concerns about academic accommodations for religious observance or religious beliefs, visit

<https://eocr.virginia.edu/accommodations-religious-observance>

or contact the University's Office for Equal Opportunity and Civil Rights (EOCR) at UVAEOCR@virginia.edu or 434-924-3200. Accommodations do not relieve you of the responsibility for completion of any part of the coursework missed as the result of a religious observance.

Safe Environment: The University of Virginia is dedicated to providing a safe and equitable learning environment for all students. To that end, it is vital that you know two values that we and the University hold as critically important:

1. Power-based personal violence will not be tolerated.
2. Everyone has a responsibility to do their part to maintain a safe community on Grounds.

If you or someone you know has been affected by power-based personal violence, more information can be found on the UVA Sexual Violence website that describes reporting options and resources available – www.virginia.edu/sexualviolence.

As your professor and as a person, know that we each care about you and your well-being and stand ready to provide support and resources as we can. As a faculty member, we are responsible employees, which means that we are required by University policy and federal law to report what you tell us to the University's Title IX Coordinator. The Title IX Coordinator's job is to ensure that the reporting student receives the resources and support that they need, while also reviewing the information presented to determine whether further action is necessary to ensure survivor safety and the safety of the University community. If you would rather keep this information confidential, there are Confidential Employees you can talk to on Grounds (See http://www.virginia.edu/justreportit/confidential_resources.pdf). The worst possible situation would be for you or your friend to remain silent when there are so many here willing and able to help.

Well-being: If you are feeling overwhelmed, stressed, or isolated, there are many individuals here who are ready and wanting to help. The Student Health Center offers Counseling and Psychological Services (CAPS) for all UVA students. Call 434-243-5150 (or 434-972-7004 for after hours and weekend crisis assistance) to get started and schedule an appointment. If you prefer to speak anonymously and confidentially over the phone, Madison House provides a HELP Line at any hour of any day: 434-295-8255.

Syllabus Note: This syllabus is to be considered a reference document that may be adjusted throughout the course of the semester to address necessary changes. This syllabus can be changed at any time without notification; it is up to the student to monitor the website for news of any changes. Final authority on any decision in this course rests with the professor, not with this document.

Research: Your class work might be used for research purposes. For example, we may use anonymized scores from student assignments to compare to other student performance data.

Any student who wishes to opt out can contact the instructor or TA to do so after final grades have been issued. This has no impact on your grade in any manner.